Bachelor's Thesis

What limits cross-lingual zero-shot-sentiment more? The writing system or resource?

Author
Rofaïda Rabehi
rofaida.rabehi@student.uni-tuebingen.de

Supervisor
Çağrı Çöltekin
cagri.coeltekin@uni-tuebingen.de

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Antiplagiatserklärung

Ich erkläre hiermit,

dass ich die vorliegende Arbeit selbständig verfasst habe,

dass ich keine anderen als die angegebenen Hilfsmittel und Quellen benutzt habe,

dass ich alle wörtlich oder sinngemäß aus anderen Werken übernommenen Aussagen als solche gekennzeichnet habe,

dass die Arbeit weder vollständig noch in wesentlichen Teilen Gegenstand eines anderen Prüfungsverfahrens gewesen ist,

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1 Introduction

Sentiment Analysis requires annotated data, which is far less available for languages outside the small group that dominates NLP research (Joshi et al., 2020). Zero-shot Classification offers a way around this were labels are assigned to a text without the model having any labeled examples for the task in its pretraining. One common approach is to to treat classification as a Natural Language Inference, short NLI, problem. (Yin et al., 2019) which allows multilingual models to be applied to languages where they have not been fine-tuned on. This is particularity useful for languages with fewer resources but this approach also has its problems. NLI models react inconsistently to minor changes in how a hypothesis is phrased seen in Arakelyan et al. (2024) and this approach has also been reported to struggle precisely for the languages it is meant to help seen in Philippy et al. (2024) with Luxembourgish.

Cross-lingual zero-shot classification also does not perform equally well across languages (Philippy et al., 2024). Two factors that may contribute this are the writing system of the target languages and the amount of data available for it. Since these two factors occur rarely independently their individual contributions are difficult to determine. Four languages were chosen for this work: German, English, Japanese and Hebrew. German, English and Japanese belong to class 5 in the taxonomy of Joshi et al. (2020), the most resourced group. [1] Hebrew is classified as a class 3, characterized by substantial unlabeled data but insufficient labeled resources and is therefore considered as the Lower-Resource Language of this work. German and English are both written in the Latin script while Japanese belongs to the same resource class as these two but uses three non-Latin scripts, Kanji, Hiragana and Katakana, which allows the writing system to be varied while holding the resource level constant. Hebrew is likewise written in a non-Latin script, the Hebrew Alphabet, which allows again the writing system to be varied but this time for a lower class language. Two models were evaluated across two domains and with three templates each. The second domain was included in order to separate domain effects from language effects. The results found didn't show measurable effect of the writing systems. The differences that were observed could be attributed to the domain, text length or the choice of mode. The remainder of this work is structures as follows: Section 2 reviews related work on multilingual encoder models, zero-shot classification and resource asymmetry between languages. Section 3 described the Datasets used in both settings and how they were prepared. Section 4 presents the models, templates and evaluation procedure. Section 5 reports the results for both settings. Section 6 is the discussion of the finding and the Limitations are in Section 7. Section 8 is the conclusion. All Materials can be found in Section 9

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	Latin Writing System	Non-Latin Writing System
High-Resource	German, English	Japanese
Lower-Resource	-	Hebrew

Table 1: Overview of the Languages

2 Background

2.1 Encoders

BERT (Devlin et al., 2019) ,Bidirectional Encoder Representations from Transformers, introduced masked language modeling, which allowed transformer models to condition on both the left and right context during pretraining. RoBERTa Liu et al., 2019 kept the architecture but revised the training procedure, showing that BERT had been under trained. The Name stands for Robustly Optimized BERT Approach. XLM-R (Conneau et al., 2019) applied the same approach

[1] https://microsoft.github.io/linguisticdiversity/assets/lang2tax.txt

to a 100 languages using CommonCrawl data. DeBERTa (He et al., 2021) introduced disentangled attention and encoding content and DeBERTaV3 He et al. (2023) refined its pretraining further. The multilingual variant, mDeBERTa, is the second model used in this work.

2.2 Zero-shot and NLI

Zero-shot classification assigns labels without any labels without any labeled examples for the task. Yin et al. (2019) formulate this as a textual entailment problem: The text to be classified serves the premise and each candidate label is turned into a hypothesis. XNLI Conneau et al. (2018) provides NLI data for 15 languages translated from MultiNLI and is commonly used to fine-tune multilingual models for this purpose. Since models like these are pretrained on far more languages than they are covered by XNLI, they can be applied to languages unseen in the fine-tuning a setting referred to as cross-lingual transfer Ebrahimi et al. (2022) However Ma et al. (2021) point out a mismatch between fine-tuning task and the inference task and Arakelyan et al. (2024) show that NLI models react inconsistently to minor changes.

2.3 Resources

3 Data

Four languages were picked for this experiment: German, English, Japanese and Hebrew. The reason for choosing these languages was due to the first three languages being High-Resource languages according to Joshi et al. (2020) , with one of them using a non Latin writing script. Japanese uses three different systems, namely Kanji, Hiragana and Katakana. For the lower-resource one, one was chosen that had enough resources for machine learning tasks and does not use the Latin script. Each language was examined in two settings: Social Media and Product Reviews, in total five Datasets as seen in Table 2.

3.1 Social Media data

The first setting is Social Media. The German dataset by Cardiff NLP (Barbieri et al., 2022) is a Twitter dataset that was originally from Cieliebak et al. (2017) but (Barbieri et al., 2022) and his colleagues compiled a unified and balanced Twitter Dataset for Sentiment Analysis where every language was equally balanced in regards to frequency. They picked eight languages, German, English,French, Spanish, Portuguese, Italian, Arabic and Hindi. Since retaining the original distribution would have skewed the corpus towards the most frequent languages, the authors capped all languages to the size of Hindi, which was the smallest one with 3033 tweets Barbieri et al. (2022). This left 1839 for training and 870 tweets for testing per language. English is from the same unified Dataset and again, same as German, was originally from a different dataset (Rosenthal et al., 2017). The purpose of putting German and English together in the first setting was due to them being from the same alphabet system, being high-resource and both of them being related as they are from the same language family.

The Japanese dataset WRIME Kajiwara et al. (2021) is an emotional intensity estimation collection in Japanese, of 17000 annotated posts from SNS, short for Social Networking Services. The original corpus is annotated with emotional intensity across eight emotions on a four point scale. A paper that was published later added sentiment polarity to it Suzuki et al. (2022) based on the average rating of three independent readers. The version that was used for this work is from said polarity layer that was converted into a binary classification task by the uploaders of the Dataset[2]. Posts were labeled as positive,negative or neutral depending on the average score , where below zero is negative, zero being neutral and above being positive. Neutral was removed in the Huggingface version and the test split used here contains 1781 examples.

The Hebrew Dataset comes from HebrewSentiment, a Sentiment Dataset from OMILAB, at the Open University of Israel (Amram et al., 2018). The Dataset consists of over 12000 comments

[2]https://huggingface.co/datasets/llm-book/wrime-sentiment

from the official Facebook page of the former Israeli President Reuven Rivlin. The data was labeled as positive, negative or neutral. The authors provided both a morpheme and token based version. The version used for this work is the token based one. The test split used here contains 2560 examples.

Setup	Language	Domain
Social Media	German	Twitter
Social Media	English	Twitter
Social Media	Japanese	SNS
Social Media	Hebrew	Facebook
Product Review	DE EN JP	Reviews
Product Review	Hebrew	Surveys

Table 2: Overview of the Datasets

3.2 Product Review data

The second setting is Product Review. Here, the three high resource languages come from the same Dataset Collection, namely the Amazon Review Multi Dataset(Keung et al., 2020). This dataset was chosen because it was a different Domain from Social Media, star ratings instead of sentiment annotations and it covers all three high-resource languages, German, English and Japanese, in one single collection thus giving us the same background for examining cross-lingual differences between the writing systems. As of 2026, the Dataset is no longer hosted by Amazon, a port was hosted on Huggingface[3] by SetFit which was used for this work. The test split contains 5000 reviews per language The test split contains 5,000 reviews per language, evenly distributed with 1000 across the five star ratings.

For Hebrew on the other hand, there was no Amazon nor a Product Review Dataset in a similar style available. The closest alternative is HebrewSentiment by the HeArabNLPProject[4]. According to the Dataset-card, it consist of short responses of surveys of different industries that were also sorted into , types. For example the Industry Food and Beverages could be of type Product, Brand or Usage and Attitude. The Sentiment of the texts were labeled as positive, negative or neutral with Annotators agreeing either with strong or weak on the Sentiment of the text. The test split contains 6510 examples. No publication documenting this collection nor the annotation is available. The Datacard claims to be from the Hebrew NLP benchmark.

3.3 Data Preparation

For the experimental setup, all datasets were reduced to a binary distinction between positive and negative, with neutral instances removed when present. In the product review setting, reviews with one or two stars were labeled negative and those with four or five stars positive. Three star reviews were excluded to keep a binary balance and not knowing where to put three stars in either positive or negative. In Social Media, the median text Length was 66 characters for German, 99 characters for English, 41 characters for Japanese and 67 characters for Hebrew. In Product Review the median was 147 characters for German, 124 characters for English, 70 characters for Japanese and 37 characters for Hebrew. This makes Hebrew the only language of the four to shrink in median size for the second setting of this work. From each dataset, 290 examples per class were taken from every language, resulting in 580 examples per language and setting. This was done so to fit the smallest corpus: After neutral was removed German and English were in test split 290 examples per class. All samples were drawn with a fixed random seed of 42. For Hebrew in Product Review, only instances with strong annotator agreement were kept, leaving 1428 Positive and 1224 Negative examples. As the social media setting is

[3]https://huggingface.co/datasets/SetFit/amazon_reviews_multi_de,
https://huggingface.co/datasets/SetFit/amazon_reviews_multi_en,
https://huggingface.co/datasets/SetFit/amazon_reviews_multi_ja

[4]https://huggingface.co/datasets/HebArabNlpProject/HebrewSentiment

three datasets were two,German and English, share one, this makes the domain less controlled. However in product review setting, German, English and Japanese share a single corpus making this domain more controlled. As mentioned earlier Hebrew had no Amazon Dataset so the closet alternative was used.

4 Method

4.1 NLI

Natural Language Inference (NLI), is a task used to infer the logical relationship between a sentence pair, called the premise and hypothesis. For example, the standard labels are entailment, neutral and contradiction. Applying Zero-Shot to it, the text thats going to be classified is the premise while each candidate label is inserted to the hypothesis template that will form a hypothesis (Yin et al., 2019). The model gives each hypothesis a score, and the label with the highest score is chosen as the prediction

4.2 Models

For this experiment, two encoder models were used and were pretrained on multiple languages and fine-tuned on Natural Language Inference: XLM-Roberta-large-xnli[5] where the model is originally from Conneau et al. (2019) and mDeberta-v3-base-xnli-multilingual.nli-2mil7[6]

XLM-R was pretrained on a 100 languages including all four languges with German with 10297M tokens (66,6 Gib), English with 55608M tokens (300.8 GiB), Japanese with 530M tokens (69.3 GiB) and Hebrew with 3399M tokens (31,6 GiB) as seen in Appendix A of Conneau et al. (2019). In the NLI version, it was fine-tuned on 15 languages including German and English but not Japanese and Hebrew. mDeBERTA is based on the DeBERTa V3 architecture He et al., 2023 and was pretrained on the CC100 multilingual dataset which covered 100 languages. mDeBERTa was fine-tuned on XNLIConneau et al. (2018) along with a multilingual NLI dataset spanning to 27 languages Laurer et al. (2022) which included both Japanese and Hebrew unlike XLM-R where only German and English were fine-tuned on NLI data.

The reason two models were picked is to see the difference between model-related effects and language effects.

4.3 Templates

Three Templates were used for this work: Default, the original Hugginface hypothesis template for the zero-shot pipeline, Sentiment_en a hard-coded English sentence that names the task and Native which was the same as Sentiment_en but each of the four languages seen in Table 3. The Comparison between Default and Sentiment_en are both in English but only Sentiment_en names the task directly. Sentiment_en and Native, in English, are the same where the english part of native acts as control

4.4 Experimental Setup

Using the Huggingface zero-shot pipeline, classification was called through said pipelines. Each text had candidate labels inserted into the hypothesis template and the model scored each resulting premise hypothesis pair. The label with the highest entailment score was taken as the prediction. For each language, the candidate labels were defined in the respective target language as seen in table 3 and mapped back to a English label for evaluation. The experiment covers the aforementioned two models, two settings, four languages and three templates. As mentioned in Data, 290 examples per class after removing neutral. All samples used a fixed

[5]https://huggingface.co/joeddav/xlm-roberta-large-xnli
[6]https://huggingface.co/MoritzLaurer/mDeBERTa-v3-base-xnli-multilingual-nli-2mil7

random seed of 42. Since the text was identical on every evaluation, the comparisons show up in pairs.

4.5 Metric

Accuracy is used as the main metric. Since all samples were balances across all classes it can be interpreted against a chance baseline of 50%. Recall is reported separately as recall positive and recall negative, to see wether a model favors one over the other. The difference between the two which was referred here as bias, indicates a direction where positive values mean that positive texts are better recognized than the negative ones and the reverse for negative bias.

Using the McNemar function from statsmodels to make Template comparisons, due to the data being in pairs and being evaluated on the same text. The test only considers so called discordant cases, meaning texts were one condition is classified correctly while the other is not. Difference where p is $< 0{,}05$ were reported as significant along with 95% confidence intervals across al accuracy values. Since the range from $\pm 0{,}02$ to $\pm 0{,}03$ anything below three % was was interpreted as not meaningful. For the Product Review Setting, the results were additionally broken down in groups of under 50, 50 to 100, 100 to 200 and over 200 due to Hebrew being considerably shorter than the other three languages in this setting. This allowed an examination if differences between languages can be explained by text length over the language itself.

Template	Hypothesis
Default	This example is { } .
Sentiment_En	The sentiment of this text is { }.
Native	en: The sentiment of this text is { }.
Native	de: Die Stimmung dieses Textes ist { }.
Native	ja: このテキストの感情は{ }です。
Native	he: ha regesh ba tekst ha ze hu { }.

Table 3: Experimental Setup for Social Media. Hebrew is in the Hebrew Alphabet in the Codesetup

5 Results

5.1 Social Media data

As in Table 4 and Table 5 they show the results for the Social Media run for both models. Following the Sentiment_en Template the Cross-Lingual Comparison goes the following: Accuracy for each Language is in both Models over the chance baseline of 50 % with English having the highest Accuracy in both, followed by Japanese, German and then Hebrew again for both. Analyzing XLM-R first the Accuracy for English is 88,3% followed by with Japanese at 84,8 %. German followed close by with 83,3% and with Hebrew being the lowest of the four with 76,7% . While mDeBERTa has the same order as XLM-R, the numbers look different: English sits at 85,7 % with Japanese at 81,2%. German is again close behind with 80,9% and while Hebrew is at last place again, it has a Accuracy of exactly 80 % . Confidence interval move from $0.026\pm$ to $0.034\pm$ through all Sentiment_en Templates for XLM-R and $0.028\pm$ to $0.034\pm$ for mDeBERTa. Differences below approximately 3 % or less are interpreted as not meaningful.

For XLM-R, five out of the six pairs of Sentiment_en are exceeding the threshold with English - Hebrew having the largest gap with 11,6 followed by Japanese - Hebrew with 8,1 points. German - Hebrew have a gap of 6,6 points with English - German being close behind with 5,0 and English - Japanese being barely over the threshold with 3,5 points and lastly German and Japanese being below the threshold with 1,5 points. For mDeBERTa the differences are substantially smaller. Out of the six pairs only half of them, namely all English pairs with English - Hebrew at 5,7 ,

English - German with 4,8 and English - Japanese at 4,5 points. The ones below were at a range between 0,3 (German- Japanese) to 1,2 (Japanese - Hebrew).

Starting with English as it achieved the highest accuracy, with XLM-R it achieved in Default 0,862, as mentioned earlier, in Sentiment_en 0,883 and the same score in Native in Accuracy making Native and Sentiment the strongest templates for English with Default being the lowest out of the three, albeit the differences between the three is at the highest point 2,1 which is deemed not meaningful.

For Japanese, the highest accuracy was achieved by Sentiment_en with 0,848 followed by Native at 0,840 and the lowest being Default at 0,778,. The difference between these three makes one not meaningful, namely the difference between Sentiment_En Native which equals to 0,8 while the gaps between Native and Default and SentimentEn and Default amount to 6,2 and 7,0 respectively where they both exceed the threshold. Japanese shows the largest template effect of all four languages.

German achieved its highest accuracy with Native at 0,848 followed by SentimentEn at 0,833 and Default at 0,809 being the lowest. The spanning between the templates here amounts to only one being over the threshold and thus being meaningful, the one being the difference between Native and Default which is at 3,9 while the others,Default and SentimentEn and Native and SentimentEN, sit at 2,4 and 1,5 respectively.

Hebrew had its highest accuracy with Native at 0,781 followed by Default at 0.778 and the lowest being SentimentEn with 0,767. The spanning between all three templates in Hebrew were under the threshold thus making them not meaningful with the difference being between 1,1 for Default and SentimentEn, Native and SentimentEn at 1,4 and Native and Default at 0,3.

To summarize the highest Template effects observed in XLM-R are seen in Japanese with two Templates at 7,0 and 6,2 and German with one at 3,9 while for English and Hebrew, the Templates made no measurable difference.

Moving now to mDeBERTa starting with English, all three templates achieve the same accuracy namely with 0,857 making it the lowest across all three spanning differences with 0. This differs with XLM-R where english was also below the threshold but was at 2,1 points

Japaneses highest accuracy is 0.822 with Native, followed by Default with 0,816 and SentimentEn with 0.812 as the lowest. The difference between all three templates are under the threshold and thus not meaningful varying between 0,4 , Default and SentimentEn, to 1,0, Native and SentimentEn, being the highest with Native and Default being somewhat in the middle with 0,6.

Germans highest accuracy sits at 0,809 which is shared by Default and SentimentEn with Native being the lowest at 0,778. The differences of both Default and Native and SentimentEn and Native sit at 3,1 making is barely over the threshold.

Hebrews highest accuracy is shared with Default and Native with 0,809 with SentimentEn being the lowest at 0,800. All different ranges of the templates are between 0 and 0,9 thus being under the threshold.

For mDeBERTa , it had technically two templates reach the threshold in German as every other language was below the threshold. But since Default and SentimentEn had the same accuracy the same effect is just observed twice. This is the only template effect observed with mDeBERTa and it barely exceeded the threshold.

Language	Template	Accuracy	Recall Pos	Recall Neg	N	Bias	Ci95
German	Default	0.809	0.872	0.745	580.0	0.128	0.032
German	Native	0.848	0.862	0.834	580.0	0.028	0.029
German	SentimentEn	0.833	0.852	0.814	580.0	0.038	0.030
English	Default	0.862	0.931	0.793	580.0	0.138	0.028
English	Native	0.883	0.890	0.876	580.0	0.014	0.026
English	SentimentEn	0.883	0.890	0.876	580.0	0.014	0.026
Japanese	Default	0.778	0.883	0.672	580.0	0.210	0.034
Japanese	Native	0.840	0.855	0.824	580.0	0.031	0.030
Japanese	SentimentEn	0.848	0.886	0.810	580.0	0.076	0.029
Hebrew	Default	0.778	0.838	0.717	580.0	0.121	0.034
Hebrew	Native	0.781	0.814	0.748	580.0	0.066	0.034
Hebrew	SentimentEn	0.767	0.831	0.703	580.0	0.128	0.034

Table 4: XLM-R Social Media

Language	Template	Accuracy	Recall Pos	Recall Neg	N	Bias	Ci95
German	Default	0.809	0.734	0.883	580.0	-0.148	0.032
German	SentimentEn	0.809	0.738	0.879	580.0	-0.141	0.032
German	Native	0.778	0.666	0.890	580.0	-0.224	0.034
English	Default	0.857	0.779	0.934	580.0	-0.155	0.028
English	SentimentEn	0.857	0.783	0.931	580.0	-0.148	0.028
English	Native	0.857	0.783	0.931	580.0	-0.148	0.028
Japanese	Default	0.816	0.745	0.886	580.0	-0.141	0.032
Japanese	SentimentEn	0.812	0.783	0.841	580.0	-0.059	0.032
Japanese	Native	0.822	0.741	0.903	580.0	-0.162	0.031
Hebrew	Default	0.809	0.748	0.869	580.0	-0.121	0.032
Hebrew	SentimentEn	0.800	0.776	0.824	580.0	-0.048	0.033
Hebrew	Native	0.809	0.721	0.897	580.0	-0.176	0.032

Table 5: mDeBERTa Social Media

Viewing Table 6 the McNemar results for the two template comparisons for each language. For XLM-R :

the comparison between Default and SentimentEn in Japanese reaches a significant outcome where $p < 0{,}001$ which also shows the highest number of discordant cases with 83. Along with Japanese Default and Native with 0,0003 and discordant cases of 94 and German Default and Native with 0,0090 and 71 cases

All other comparisons range from $p = 0{,}0933$ to $p = 0{,}5510$.

For mDeBERTa, two template comparisons of German are significant with Sentiment_En and Native at 0,0072 and a discordant case of 40, along with Default and Native with a case of 56 discordant, the renaming comparison ranging from $p = 0{,}4173$ to 0,8828. The English comparison between SentimentEn and Native could not be tested due the conditions being identical which also resulted in the zero discordant cases.

In both models one template was significant which was German Default and Native where XLM-R had the higher accuracy for this where Default is 0,809 , which is the same as mDeBERTa, but Native being 0,848 whereas its 0.778 for the latter model.

The biases show for XLM-R a positive outcome where the lowest one was 0,014 for English Native and Sentiment_en to the strongest 0,210 in Japanese Default. mDeBERTa on the other hand has a negative bias for all templates where range goes from -0,048 in Hebrew Sentiment_en to the strongest one which is German Native with -0,224. While XLM-R recognizes positives

text more reliably, mDeBERTa does the same but for the negative classes across all languages and templates.

With XLM-R, the bias is largest under default in every language except Hebrew where it is just 0,007 points behind Sentiment_en. Sentiment_en and Native reduce the bias substantially in English. It drops from 0,138 to 0,014 while for Japanese its drops from 0,210 in Default to 0,076 with Sentiment_en and 0,031 with native. With mDeBERTa no comparable pattern was observed. Sentiment_en reduces the bias for Japanese and Hebrew but had little effect for German and English, while native increases in it for three languages.

Language	Comparison	XLM-R	XLM-R Disc.	mDeBERTa	mDeBERTa Disc
German	default vs sentimentEn	0.0933	60	0.8676	36
German	default vs native	0.0090	71	0.0231	56
German	sentimentEn vs native	0.1763	35	0.0072	40
English	default vs sentimentEn	0.1757	66	0.8445	26
English	default vs native	0.1757	66	0.8445	26
English	sentimenEn vs native	0	0	0	0
Japanese	default vs sentimentEn	$<$0.001	83	0.8828	46
Japanese	default vs native	0.0003	94	0.6265	38
Japanese	sentimentEn vs native	0.5510	45	0.4173	38
Hebrew	default vs sentimentEn	0.5186	60	0.5108	37
Hebrew	default vs native	0.8989	62	0.8852	48
Hebrew	sentimentEn vs native	0.1859	28	0.5510	45

Table 6: P Value and McNemar Results for Social Media. Disc. denountes the number Discordant cases

5.2 Second Setting

In Table 7 and 8 the results for the Product Review can be observed: All languages are above the chance baseline of 50 % . Hebrew achieves the highest score out of all four languages across all templates in both models which reversed the ordering seen in the social media setting where it was the lowest and now the highest. Using Sentiment_en again to analyze XLM-R , as mentioned, had its highest score with Hebrew Sentiment_en with 90,3 following language wise, the next highest score was in German with 86,6% then English with 85,7% and then Japanese with 84,5% . Confidence interval move from 0,023± to 0,031± through all Sentiment_en templates in XLM-R.

For XLM-R, three out of the six pairs are exceeding the threshold with all of them involving Hebrew: Hebrew-Japanese having the largest difference with 5,8 points, followed by Hebrew-English with 4,6 points and then Hebrew-German with 3,7 points being the lowest. The remaining three are all below the threshold with German-Japanese being 2,1 followed by English-Japanese with 1,2 and German-English being the lowest with 0,9. This makes all three Hebrew pairs meaningful.

Moving to mDeBERTa similar is observed: Hebrew is the highest with 0,922 in Sentiment_en , following English 87,8% , German with 86,2% and the lowest 82,2%. Confidence intervals

Confidence intervals move from 0,021 ± too 0,031 ± through all pairs in mDeBERTa. Five out of the six pairs exceed the threshold, namely Hebrew-Japanese having the biggest difference with 10,0 points, followed by Hebrew-German with 6,0, English-Japanese with 5,6 , Hebrew-English with 4,4 and the lowest out of all meaningful observations was German-Japanese with 4,0. Only German-English remains below the threshold with 1,6 points.

Going now to the template comparison, XLM-R shows only small differences in the product review setting. For Hebrew, Native achieves the highest accuracy with 91,4%, followed by Sen-

timent_en with 90,3% and Default with 88,1%, a gap of 3,3 points which exceeds the threshold. For the other three languages the gap remains below it as English varies by 0,4 points, German by 1,2 and Japanese by 1,9.

For mDeBERTa, Japanese shows the only template effect above the threshold. Native achieves 86,2% followed by Default with 85,3% and Sentiment_en being the lowest at 82,2%. The gaps between Native and Sentiment_en which are 4,0 points, and between Default and Sentiment_en which are 3,1 points, exceed the threshold while Native and Defaults difference is only 0,9. For the other three languages the gaps remain below the threshold with German varying by 1,7 points , Hebrew by 0,9 points and English by 0,8 points.

Viewing Table 9 the McNemar results for the two template comparisons for the Product Review Setting: For XLM-R there is only one comparison that reached significant result that being Hebrew Default and Native with p = 0,0171 and 57 discordant cases. The remaining comparisons range from p = 0,0673 to p = 0,9121. For mDeBERTa, two comparisons are significant and they are both in Japanese: Sentimen_en vs Native with p = 0,0006 and 41 Discordant cases and Default vs Sentiment_en with p = 0,0072 and 40 Discordant cases. All other comparisons range from p = 0,1748 , German Default vs Sentiment_en, to p = 0,8638 with Hebrew default vs Sentiment_en. Unlike the social media setting where German Default vs Native was significant in both models, no comparison in the Product Review Setting reached significance in both. The English comparison between Sentimen_en and Native could not be tested as the condition is identical

Moving now to the last section, Bias shows in XLM-R again positive bias with the strongest bias being 0,210 in Hebrew Default and Japanese Default. No negative was observed in XLM-R. In mDeBERTa it reverses the pattern seen in the social media setting where the ten out of the twelve conditions show a positive bias, going from 0,014 for German Default to 0,197 for Japanese Sentiment_en. The two negative Biases are from Hebrew with -0,010 for Hebrew Sentiment_en to -0,041 in Hebrew Native.

Language	Template	Accuracy	Recall Pos	Recall Neg	N	Bias	Ci95
German	Default	0.859	0.934	0.783	580.0	0.152	0.028
German	Native	0.871	0.903	0.838	580.0	0.066	0.027
German	SentimentEn	0.866	0.903	0.828	580.0	0.076	0.028
English	Default	0.853	0.952	0.755	580.0	0.197	0.029
English	Native	0.857	0.924	0.790	580.0	0.134	0.028
English	SentimentEn	0.857	0.924	0.790	580.0	0.134	0.028
Japanese	Default	0.826	0.931	0.721	580.0	0.210	0.031
Japanese	Native	0.829	0.897	0.762	580.0	0.134	0.031
Japanese	SentimentEn	0.845	0.907	0.783	580.0	0.124	0.029
Hebrew	Default	0.881	0.986	0.776	580.0	0.210	0.026
Hebrew	Native	0.914	0.972	0.855	580.0	0.117	0.023
Hebrew	SentimentEn	0.903	0.983	0.824	580.0	0.159	0.024

Table 7: XLM-R Product Review

Language	Template	Accuracy	Recall Pos	Recall Neg	N	Bias	Ci95
German	Default	0.879	0.886	0.872	580.0	0.014	0.027
German	Native	0.874	0.897	0.852	580.0	0.045	0.027
German	SentimentEn	0.862	0.921	0.803	580.0	0.117	0.028
English	Default	0.886	0.903	0.869	580.0	0.034	0.026
English	Native	0.878	0.921	0.834	580.0	0.086	0.027
English	SentimentEn	0.878	0.921	0.834	580.0	0.086	0.027
Japanese	Default	0.853	0.900	0.807	580.0	0.093	0.029
Japanese	Native	0.862	0.900	0.824	580.0	0.076	0.028
Japanese	SentimentEn	0.822	0.921	0.724	580.0	0.197	0.031
Hebrew	Default	0.922	0.931	0.914	580.0	0.017	0.022
Hebrew	Native	0.931	0.910	0.952	580.0	-0.041	0.021
Hebrew	SentimentEn	0.922	0.917	0.928	580.0	-0.010	0.022

Table 8: mDeBERTa Product Review

Language	Comparison	XLM-R	XLM-R Disc.	mDeBERTa	mDeBERTa Disc
German	default vs Sentiment_en	0.6714	50	0.1748	44
German	default vs native	0.4185	55	0.7488	39
German	Sentiment_en vs native	0.6892	25	0.2652	29
English	default vs Sentiment_en	0.8937	56	0.5218	39
English	default vs native	0.8937	56	0.5218	39
English	Sentiment_en vs native	0	0	0	0
Japanese	default vs Sentiment_en	0.2004	61	0.0072	40
Japanese	default vs native	0.9121	82	0.4862	33
Japanese	Sentiment_en vs native	0.2002	39	0.0006	41
Hebrew	default vs Sentiment_en	0.0673	43	0.8638	34
Hebrew	default vs native	0.0171	57	0.4862	33
Hebrew	Sentiment_en vs native	0.3074	24	0.3827	21

Table 9: P Value and McNemar Results for Product Review. Disc. denotes the number Discordant cases

6 Discussion

6.1 Writing System

The experiments show that there was no disadvantage for Japanese, the high-resource non-Latin writing script. In the Social Media setting, the difference between German and Japanese stays below the threshold both models with 1,5 points for XLM-R and 0,3 with mDeBERTa. Compared to English, the difference exceeds the threshhold with 3,5 points for XLM-R and 4,5 for mDeBERTa. This is notable because Japanese was not covered in the fine-tuning data of XLM-R and has the smallest token count of all four languages, although the corpus size of Japanese exceeds that of German, by 2,7 GiB, which hints that token counts are not directly comparable across writing systems.

6.2 Resource Level

While Hebrew was the lowest on the first setting 76,7% it was the best scoring language in the second setting 90,3%. This reversal trend is not language based though: the Hebrew Product Review Dataset uses a different type of text along with a shorter median length of 37 characters compared to 67 in the social media setting and contains only 473 unique texts among the 580 examples. The resource effect cant be separated from text length and domain.

6.3 Model dependecy

XLM-R shows a positive bias across all conditions in both settings, while mDeBERTa is entirely negative in social media and mostly positive in product reviews. The only template effect that was significant in both models is the German Default vs Native, but it points in the opposite direction where in XLM-R, Native is 3,9 points higher than Default but in mDeBERTa its the other way around with 3,1 points lower. This hints that the template effect is model based rather than a property of the approach. Another difference between the models is mDeBERTas NLI fine-tuning data covered Japanese and Hebrew, while XLM-R had only German and English of the four target languages.

6.4 Templates

Native showed no consistent advantage over Sentiment_En, not even for Japanese and Hebrew, which were not covered by the NLI fine-tuning data of XLM-R. Across both models and settings, the different ranges from -3,1 points to 4,0 points. Only two values exceed the threshold: German with mDeBERTa in Social Media, where Native is 3,1 pointers lower and Japanese also with mDEBERTa in Product Reviews, where it is 4,0 points higher. Both exceptions occur in the same model and point in opposite directions.

6.5 Consequence

The experiment started with a clear language difference. Hebrew performed the worst in the Social Media setting but ranked the highest in the Product Review Setting. The Bias direction reversed between models and between settings. Out of all template effects, none was significant in both models in the same direction. After controlling for text length, model and domain, little of the initial difference remained. Cross-lingual comparisons that do not control for these factors risk attributing differences to language that in fact stems from domain, text length or model choice.

7 Limitations

7.1 Domain

In the Social Media Setting, all four languages came from separate collection effort with different annotation procedures. Language and domain are therefore confounded thus making the observed differences between the languages in this setting not be attributed to language alone. Only the comparison between German, English and Japanese in the Product Review setting is truly domain controlled.

7.2 Hebrew

There is no public available Product Review Dataset for Hebrew. Mughaz et al. (2018) describes a corpus of cellphone reviews in Hebrew for Sentiment Analysis but this dataset is not publicly available. The Dataset that was used instead consist of short survey responses and is neither accompanied by a publication documenting the collection nor the annotation procedure. The resource comparison is here too confounded with the domain.

7.3 Text length

The corpora differ in median text length , ranging from 37 to 147 characters. The length control applied in the product review setting mitigates this but some language-length groups contain very few examples where there only one single text. To add to that, the Hebrew Product Review sample contains only 473 unique texts among the 580 examples, which means the effective sample size for this setting is smaller than reported.

7.4 Sample size

Each condition was evaluated on 580 examples, resulting in confidence intervals of approximately three percentage points. Many of the observed difference fall below this threshold and can not be interpreted.

7.5 Model

Only two models were observed, both of them which use an NLI zero-shot approach. Other zero-shot methods like prompting with generative models or masked language modeling were not considered and the findings found in this work may not necessarily transfer to them.

7.6 Binary Classification

Neutral labels were removed from all datasets, as the class was either absent or too small in two of the Datasets. Sentiment classification with three labels is more difficult and the results reported here do not transfer to that setting.

8 Conclusion

This work examined if the writing system or the resource level of a language limits cross-lingual zero-shot sentiment classification more. Four languages were evaluated across two settings with two models and three templates each. No measurable effect of the writing system was found. Japanese, which uses three non-Latin scripts, remained indistinguishable from German in the social media setting with both models, despite not being covered by the NLI fine-tuning data of XLM-R. The effect of the resource level did not hold either. Hebrew performed the worst in the social media setting but best in the product review setting. This reversal could be traced to domain, duplicate texts and text length rather than to the language itself. What did emerge however was the extent to which results depend on factors other than language. Bias direction reversed between models and between settings and no template effect was significant in both models in the same direction as Native was better in XLM-R and worse in mDeBERTa. Cross-lingual comparisons need to control for domain, text length and model choice or else the differences from these sources look like language effects. Future work could extend to a lower-resource language written in the Latin script and other zero-shot approaches.

9 Material

The Codeframework can be found on Github

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A Use of AI tools

Generative AI (Claude) was used in debugging the code in preparation of the Thesis. All Analysis and result interpretations were made by the author.